

Can entertainment and information gratifications influence compulsive short video app use? The pivotal function of immersive experience and emotional attachment

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ABSTRACT

Existing studies have documented the detrimental consequences associated with the compulsive and excessive utilization of short video app, yet the exact causes of these problematic usage patterns remain unclear. Addressing this gap, this study employs the Stimulus-Organism-Response (SOR) paradigm to investigate how entertainment, information, and sociality gratifications influence immersive experience and emotional attachment, and how these psychological states in turn drive compulsive usage behaviors among young users. Employing a questionnaire survey method to collect data, the research uses a sample of 896 short video app users and employs structural equation modeling to validate a conceptual framework. The findings reveal that entertainment gratification and information gratification exert a significant influence on users' immersive experience and emotional attachment. Furthermore, this study discovers that immersive experience and emotional attachment serve as two mediators in shaping the influence of users' gratifications sought on compulsive app utilization. By identifying the factors driving compulsive use among younger generation, this research provides insights for app developers and organizations within the short video app industry, as well as for public health authorities and educational institutions.

1. Introduction

Recently, the proliferation of short video applications (apps) has enabled users to generate and disseminate content spanning seconds to minutes, which has surged in popularity unprecedentedly [1,2,3]. Their meteoric ascent in popularity, driven by captivating formats, distinct thematic categories, expeditious tempo, tailored content suggestions, and the pervasive influence of entertainment figures, has eclipsed conventional social media platforms [4,5]. Unlike traditional mobile apps, short video platforms like TikTok provide a convenient way for users to create, edit, and share videos [2]. Recent statistics show that, a notable 67 % of short video app users within the age bracket of 18–19 exhibited active engagement with the platform, while an estimated 56 % of individuals aged 20–29 displayed a comparable level of utilization [6]. The demographic of young people has emerged as the primary driving force behind mobile internet and short video sectors [7,8]. Their engagement peaks during daily moments of boredom, as they seek experiential stimuli by interacting with user-generated content in short video apps, a process that ultimately immerses them in these captivating

experiences [9,5].

The swift rise of short video industry, exemplified by platforms such as TikTok, marks a transformative phase in media consumption [10]. Short video consumption has become a key leisure activity, as its high information density and fragmented, mobile nature efficiently fill users' spare time, enabling quick information absorption [11]. Research has indicated that short video apps employ the mechanism of personalized recommendation through comprehensive analysis of big data, effectively tailoring content of interest to users and engrossing them within immersive experience, thereby fostering a propensity for compulsive usage [12,13,14]. Recent scholarship further underscores that algorithmic curation, such as TikTok's clustered publics prioritizing interest-based content aggregation, intensifies compulsive engagement by amplifying immersive cycles, while opacity in recommendation architectures further hamstrings users' self-regulatory capacities [15,16].

Compulsive short video app use constitutes a problematic online behavior, characterized by excessive time investment and loss of control [17]. It differs from addiction, which involves not only compulsive behavior but also physical or psychological dependence, including

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cravings, tolerance, and withdrawal symptoms [6]. Thus, while compulsive use may feature in addiction, it alone does not suffice to diagnose an addictive disorder unless significant harm and dependency are evident [18]. Drawing on existing literature, this article delineates compulsive use of short video apps as a condition wherein users exhibit an incapacity for self-regulation, such as an inability to govern the frequency or duration of their interactions on the platform. Numerous young adults have developed such compulsive behaviors, negatively impacting their academic success, social relationships, career progression, and compromised physiological and psychological health [14].

Previous academic research on compulsive use of apps like TikTok and Instagram [9,5] draws parallels with other social media addictions, linking overuse to emotional exhaustion [19] and highlighting personality traits as mediators [20]. Recent research has demonstrated the significant role of social and prosocial factors in contributing to compulsive use behaviors on these apps [7]. Collectively, these studies contribute to understanding how various aspects of short video platforms influence compulsive use, offering valuable insights into the mechanisms underlying these behaviors and potential pathways for intervention. Despite these advancements, there is a notable dearth of attention regarding the psychological mechanisms underlying the development of compulsive use behaviors among users of short video app.

To investigate compulsive short video app use, understanding why users perceive these platforms as indispensable is a prerequisite. This requires examining the multifaceted benefits they offer. The uses and gratifications theory explains the satisfactions driving sustained engagement with short video apps [21,6], clarifying media consumption patterns and motivations for prolonged engagement. Furthermore, from a more profound perspective concerning the psychological aspects of user experience, the academic community has examined two closely associated factors, namely immersive experience and emotional attachment, which are intricately linked to user engagement behaviors [1,22,8]. While previous research has investigated some factors influencing compulsive usage of short video apps, a comprehensive examination of the underlying psychological mechanisms remains lacking. Specifically, the mediating roles of immersive experience and emotional attachment in linking user gratifications to compulsive behaviors require further investigation. Given these gaps, the current research aligns well with the Stimulus-Organism-Response (SOR) framework, as it elucidates how external gratifications sought (Stimulus) influence internal psychological states (Organism), ultimately driving compulsive behaviors (Response). Additionally, few studies to date have integrated the SOR model specifically to examine compulsive usage, suggesting an opportunity for a more nuanced application of this framework. Finally, there has been limited research on compulsive usage patterns among young users, who often struggle with self-control and are particularly prone to becoming addicted to short video apps [1]. Their susceptibility makes this demographic critical to study. Consequently, the present study endeavors to address the following research questions:

RQ1: Do perceived gratifications derived from short video app use directly catalyze immersive experience and foster emotional attachment?

RQ2: Do immersive experience and emotional attachment directly stimulate compulsive short video app use?

RQ3: Do immersive experience and emotional attachment act as mediators in the association between gratification and compulsive short video app use?

The resolution of the issues will be necessary in the contemporary digital network environment. First, this study focuses on the behavioral patterns of young users engaging with short video apps, aiming to explore the psychological factors that drive their continuous viewing behavior. As a demographic highly susceptible to app addiction, their behavioral motivations warrant societal attention. Second, the study develops a framework, grounded in the SOR model, that identifies both the immediate and underlying factors driving compulsive behavior in

short video app usage, as well as key mediators involved in this process. By exploring the hypothesized association and pathways, this research has the potential to supplement existing investigations that explore the interconnections among gratifications sought, immersive experience, emotional attachment, and compulsive usage of short video apps. Lastly, findings offer practical insights for app developers, public health authorities, and other stakeholders to design interventions promoting a healthier digital ecosystem.

2. Theoretical framework and hypotheses development

2.1. Linking gratifications sought to immersive experience

The delineation of immersive experience serves as the fundamental construct employed to encapsulate mobile interactions within the purview of research across communication, marketing, digital gaming, and cognate disciplines [23,24]. Scholars have recognized immersive experience as a pivotal dimension of positive psychology, underscoring its interdisciplinary relevance [25,26], and it is widely applied in domains such as short video, online gaming, virtual environments, and educational settings [27]. Immersive experience is defined by deep engagement, intrinsic enjoyment, and a seamless connection between external stimuli and individual responses [28,23]. When immersive experience occurs, the focus is placed on the specific activity under consideration, with any external disturbances deemed irrelevant, hence blurring the distinction between the actor and the action itself [8]. The combination of visually engaging or aesthetically pleasing visuals and accompanying music in short video s creates an immersive experience for users. As users become absorbed in concise video s, subsequent immersion amplifies the effects of positive experiences, fostering an increased inclination to persist in their utilization. Users may display diminished interest in their immediate surroundings, perceiving their mundane offline existence, thus fostering a reliance on short video apps [1,6]. While immersive experience and flow are conceptually related, they represent distinct psychological constructs. Flow reflects a state of profound concentration wherein an optimal balance between an individual's skills and the presented challenges results in maximal engagement [27]. Conversely, immersive experience emphasizes broader sensory absorption, even if the task is less skill-challenging. Both constructs share characteristics such as intense focus and diminished self-awareness, however, flow uniquely requires the congruence of skill and challenge, whereas immersion does not. Immersive experience can be considered a precursor to the flow state, fostering a pleasurable cycle of engagement devoid of extrinsic rewards [23]. In terms of measurement, while flow is operationalized through scales emphasizing skill-challenge balance and task absorption, immersive experience is measured via items capturing sensory engagement and temporal dissociation, irrespective of task difficulty [13].

Gratifications sought derived from entertainment, information, and sociality address users' psychological and social needs, driving engagement with short video apps [21,29,30]. Entertainment gratification addresses the intrinsic desire for pleasure and stress relief, enhancing emotional connection and promoting sustained use. Information gratification fulfills users' need for knowledge acquisition and cognitive stimulation, which is especially significant in environments characterized by rapid content updates. Meanwhile, sociality gratification facilitates a sense of community and fosters user interaction and social engagement, both of which are key elements contributing to the app's overall appeal. Entertainment gratification refers to hedonic enjoyment, stress reduction, autonomy, and satisfaction of socio-psychological needs through communication technology use [31,32]. Information gratification delineates the inherent inclination of individuals to actively pursue and acquire information as a means to satiate their innate curiosity and augment their understanding of the surrounding milieu [33]. Sociality gratification involves social presence and interaction, which have been linked to digital media usage

[25,34,35]. Collectively, these three types of gratification serve as driving forces that contribute to individuals' immersive experience in digital environments, including short video apps, social media, and online games. An immersive experience is characterized by captivating emotional interaction, which aligns closely with these forms of gratification.

Research on how gratifications influence digital media use shows that entertainment gratification, driven by hedonistic desires, significantly shapes engagement with intelligent media [30,36]. This desire for entertainment is particularly relevant in immersive environments, where the principal characteristic is the ability to provide captivating and enjoyable experiences. The emotional interaction between users and the medium forms the core of this immersive experience. Empirical studies confirm that entertainment gratification has a strong influence on users' online immersion [37,38]. In addition to entertainment gratification, information gratification plays a crucial role in motivating media usage, particularly in contexts involving mobile devices and short video apps, where the rapid acquisition of knowledge is a prominent feature [39,4]. Consequently, this hypothesis suggests that when individuals are immersed in the endeavor of discerning relevant information amidst a voluminous dataset, they exhibit a diminished awareness of temporal passage. This cognitive absorption contributes to immersive experiences by linking user engagement with information processing [33]. Furthermore, sociality gratification has been identified as another key factor that influences digital media use, often through social interactions such as liking, commenting, and sharing [40]. Multiple studies on social network services consistently demonstrate that sociality gratification significantly influences various dimensions of immersive experience [21,38]. For instance, social interaction in online games generates flow, indicating a positive correlation between sociality gratification and immersive experience in computer-mediated environments [24]. Moreover, research suggests that these social interactions not only enhance users' immersion but also contribute to the development of habitual behaviors, which in turn lead to increased consumer satisfaction [41,27]. Considering the discussions, the subsequent hypotheses are posited:

H1: Entertainment gratification is positively associated with immersive experience.

H2: Information gratification is positively associated with immersive experience.

H3: Sociality gratification is positively associated with immersive experience.

2.2. Linking gratifications sought to emotional attachment

Emotional attachment can be understood as a concept rooted in interpersonal connections, which signifies the emotional linkage forged between an individual and a consumer entity, such as a brand, person, location, or product [42,43,44]. Attachment studies widely recognize that emotional attachment to a specific object is a paramount human requisite [45]. Emotional attachment significantly influences motivation and behavior, as individuals who develop emotional ties to a product tend to invest time and effort to strengthen that connection [46]. This correlation has recently been used to clarify emotional attachment's integral role in driving individuals' deep engagement with social media and short video apps [47,48].

Antecedent research has postulated that entertainment gratification assumes a pivotal role in delineating the extent of individuals' engagement with social media platforms [49,32]. Building upon this premise, scholars have identified a positive relationship between perceived entertainment gratification and emotional attachment [34,50]. An individual who derives pleasure from engaging with social media platforms may develop a sense of dependency, experiencing emotions akin to an inability to function without it. In this scenario, individuals are inclined to show greater psychological engagement and cultivate

emotional attachment to their used social network [51]. Regarding short video apps, users' intrinsic creativity and enthusiasm are leveraged to foster the dissemination of a diverse array of compelling video content, which in turn cultivates entertainment and leisure opportunities [52]. In contemporary society, where individuals often grapple with the vicissitudes of life and the associated stressors, the entertainment offerings of short video apps have emerged as a source of comfort, thereby engendering a robust emotional bond between the app and its user base [26,53].

Additionally, prior studies confirm that aligning knowledge accuracy with user expectations positively influences participants' emotional states in community discussions [42,33]. Task-technology fit theory posits that a platform's effectiveness hinges on its capacity to fulfill user needs. Short video platforms have introduced a variety of features that allow users to share and access information in diverse formats, including text, images, and videos [39]. The platform's functions effectively fulfill users' needs and contribute to their acquisition of informational gratification. As users utilize the platform's resources to access information through short video s, they develop a connection to its features. Therefore, the gratification gained from engaging with the informational content of short video s fosters emotional attachment to the platform [47,44].

Furthermore, when participating in online interactions on short video app, individuals create virtual identities and cultivate connections to obtain sociality gratification, which encompasses emotional support and interpersonal closeness [22,54]. Berger pointed out that fostering emotional bonds within social context is facilitated through infusing the means of interaction with a strong sense of sociability [55]. Anchored in attachment theory, relationships are delineated as fundamental human necessities, with positive and benevolent interactions acting as the bedrock for the formation of emotional bonds [42]. It can be inferred that users who forge close-knit relationships within their short video ecosystems are likely to be more proactive in their usage of the platforms, while experiencing heightened emotions that include tenderness and recognition. Consequently, individuals are anticipated to manifest a heightened proclivity towards psychological engagement, a diminished sense of apathy towards short video app usage, and an augmented likelihood of forming emotional attachment [21,44]. This discussion lays the groundwork for the postulation of the ensuing hypotheses:

H4: Entertainment gratification is positively associated with emotional attachment.

H5: Information gratification is positively associated with emotional attachment.

H6: Sociality gratification is positively associated with emotional attachment.

2.3. Linking immersive experience and emotional attachment to compulsive short video app use

Compulsive use refers to persistent engagement with technological platforms driven by repetitive and frequent behaviors, despite users recognizing potential negative consequences [56,40,9]. Neurobiological alterations, both in functionality and structure, analogous to those noted in other forms of addiction, have been identified in individuals compulsively utilizing social media platforms [57]. These changes may instigate a compelling urge to indulge in social media, intensifying cravings and culminating in behaviors focused on seeking satisfaction. In the contemporary milieu, short video apps have sparked a global phenomenon, serving as a powerful medium for individuals to create and share concise video s, typically around one minute in length. Given the inherent satisfaction derived from both the creation and consumption of succinct audio-visual content, individuals are progressively susceptible to succumbing to the allure of obsessive and compulsive utilization of short video apps [5,53].

The establishment of immersive experience is designed to facilitate users' continual exploration and interaction with online content. In

computer-mediated environments, achieving immersion is particularly critical for sustaining user engagement [58,24]. Previous research has demonstrated that the experience of flow exerts a positive influence on learning outcomes, mindsets, desires, and actions [28]. However, it has been contended that the positive experience of immersion may serve as a plausible precursor to compulsive behaviors. For instance, recent studies indicate that individuals seeking to maintain a positive immersive experience while using social media apps on their cellphones are more susceptible to acquiring detrimental addictive habits [35]. Furthermore, within the domains of digital gaming, websites, as well as online communities, antecedent inquiries posited that undergoing an immersive experience has the possibility to instigate addictive conduct and compel individuals toward compulsive involvement with social media platforms [39,13,40].

Attachment theory suggests that individuals who form emotional bonds with short video platforms are more likely to spend extended time and effort engaging with digital environments, which can increase the likelihood of compulsive short video app utilization [26,48]. The significance of emotional attachment in fostering prolonged user interaction with websites has been well-documented [34,22]. Similarly, research has demonstrated that affective bonds within online communities significantly enhance user engagement and retention [49,44]. Furthermore, mood modulation is a key factor in the transition from regular to compulsive smartphone use, with a direct correlation to compulsive behaviors [12]. Short video apps, with features like filters and visual enhancements that improve video quality, encourage greater user engagement [1,4]. As users increasingly rely on these platforms to create and share video s, habitual dependencies form, fostering stronger attachment and frequent use, a progression that increases the likelihood of compulsive use. This progression increases the likelihood of compulsive use. Accordingly, the following hypotheses are proposed:

H7: Immersive experience is positively associated with compulsive short video app use.

H8: Emotional attachment is positively associated with compulsive short video app use.

3. Research methodology

3.1. Research model

In accordance with the stimulus-organism-response (SOR) framework, external ecological stimuli serve as triggers that incite an internal cognitive and affective state, ultimately manifesting in corresponding behavioral responses [40]. The SOR framework intricately delineates consumer behavior by elucidating three pivotal constituents: stimulus (comprising the motivational factors for engagement), organism (encompassing the emotional and affective responses elicited by the stimulus, subsequently cognitively processed by consumers), and response (pertaining to the actions undertaken by consumers). This model has demonstrated its applicability across various spheres,

encompassing virtual communities [29] and digital commerce [59] in the prevailing circumstances Fig. 1.

The present study employs the SOR framework as its theoretical foundation, a choice that is justified by its unique capacity to address the study's objectives more comprehensively than alternative theoretical approaches. Unlike frameworks that focus primarily on technological acceptance mechanisms, user motivations, or cognitive learning processes, the SOR framework offers a dynamic and hierarchical model to analyze the sequential relationship between environmental stimuli, internal psychological states, and behavioral outcomes. Specifically, the exogenous stimuli encompass entertainment gratification, information gratification, and sociality gratification, while the internal organisms encompass immersive experience and emotional attachment. The response in this study refers to compulsive use of short video apps by young users.

Given its proven efficacy in analogous research contexts, we employed this model to ensure a robust and validated approach for examining the complex interplay between sought gratification stimuli, psychological states, and user behaviors. Moreover, the SOR model's emphasis on the organism's internal state as a mediator aligns seamlessly with the study's focus on probing the cognitive and emotional processes underlying compulsive behavior. This investigation is particularly salient in light of the study's focus on variables such as emotional attachment and immersive experiences, which are directly shaped by the stimuli offered by the applications and, consequently, influence users' behavioral responses in terms of engagement patterns.

3.2. Sample and data collection

The focal demographic of this investigation comprises youthful users in China who use TikTok (called Douyin in the Chinese market). TikTok, being one of the preeminent globally recognized platforms for short video content, has gained substantial traction in the Chinese market [7,48]. Given its early entry into the Chinese market, the large user base of TikTok in China constitutes an ideal cohort for comprehensively examining short video usage patterns. Consequently, Chinese TikTok users present themselves as a highly representative sample for scrutinizing the phenomenon of compulsive short video app use. The demographic focus on young users aged 18–33 within China was strategically chosen due to their high engagement rates with short video apps, making them a pivotal group for understanding the dynamics of compulsive usage. This age range was selected because it denotes a key demographic with frequent engagement in short video platforms, offering valuable insights into the patterns and consequences of compulsive usage within this group. Simultaneously, in previous studies, scholars have often regarded the 18–33 age range as a crucial demarcation for the young adult demographic [60,61]. To explore the aforementioned associations, the research meticulously devised a survey to gather empirical evidence. In conducting this study, meticulous attention was paid to ethical considerations, especially given the sensitivity of

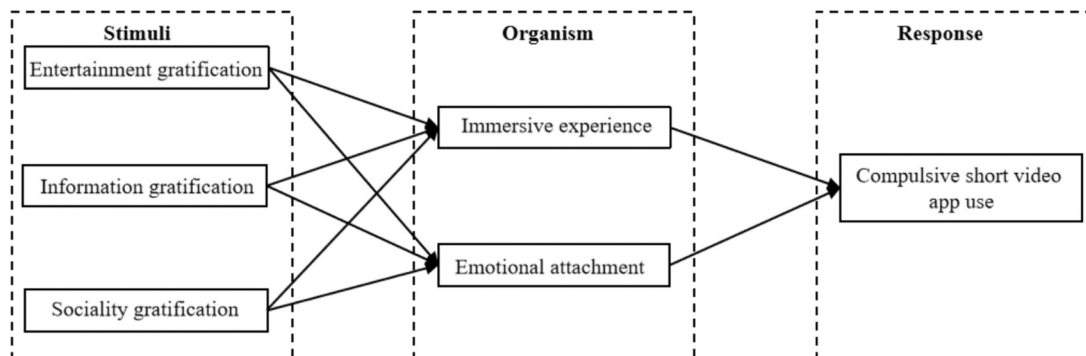


Fig. 1. The conceptual research model.

exploring compulsive behaviors akin to addiction. Initially, this research executed the recruitment of participants with utmost care, ensuring explicit authorization was secured from all individuals. Participants received thorough briefings regarding the research’s objectives, the essence of the questions, and their entitlement to disengage at any juncture without repercussions. Moreover, the research protocol, including participant recruitment and data handling procedures, secured endorsement from the institutional review board (IRB), confirming adherence to ethical standards in research involving human subjects.

To guarantee the collection of scientifically robust and methodical information, a pretest was initially conducted to solicit crucial feedback. Initially, a set of 50 questionnaires was disseminated, and subsequent deliberations were undertaken based on the received feedback, leading to the refinement of the questionnaires. The ultimate rendition of the survey, composed in the Chinese language, was uploaded to the widely-utilized online survey platform known as Sojump to facilitate participants’ accessibility. The online investigation spanned a duration of three weeks, during which the expected number of questionnaires was completed. Ultimately, the study amassed a dataset of 896 questionnaires, deemed suitable for rigorous analysis, which will be employed in further research. The sample size was determined based on the guideline that a minimum of 10–20 participants per observed variable is recommended for structural equation modeling, with our final sample exceeding this threshold to ensure statistical power [62].

3.3. Measurement

The questionnaire for this study consisted of two sections. The initial segment included five demographic variables: gender, age, educational background, students’ experience with short video app usage, and their daily time allocation towards short video apps. The second segment involved six variables, as elaborated upon in the section on theoretical framework and hypotheses development. The selection of measurement items for these six parameters was meticulously conducted, drawing from existing scholarly sources to ascertain their validity and reliability. This study specifically modified some items to align with the unique investigative context.

The quantification of compulsive use of short video apps involved the compilation of six items, which underwent requisite adaptations derived from the empirical investigation conducted by Choi and Lim [63]. The assessment of emotional attachment encompassed four indicators, appropriately modified based on the scholarly inquiry conducted by Thomson et al. [45]. The evaluation of immersive experience comprised a quartet of dimensions, suitably adapted from the scholarly endeavors of Lu and Cheng [13]. The scales of information gratification (comprising three indicators) and sociality gratification (comprising four indicators) underwent meticulous modifications informed by the study performed by Patwardhan et al. [64]. The measurement of entertainment gratification was refined by an empirical investigation undertaken by Hsu et al. [31], which included a set of three items. The entire set of 23 items was evaluated using a five-point Likert scale, spanning from ‘strongly disagree = 1’ to ‘strongly agree = 5’. Table 1 delineates the inventory of measurement indices along with their corresponding sources.

Given that the original items were in English and data collection took place in China, back-translation was used to verify consistency between the English and Chinese versions. Furthermore, a panel of 5 academics from the field of new media was convened to scrutinize any ambiguous phrasing, thereby substantiating the questionnaire’s validity prior to its dissemination. In light of the findings, certain modifications were implemented to enhance the precision and comprehensibility of the scale. Subsequently, a pilot test was conducted with 50 preliminary participants to assess item clarity, response feasibility, and measurement reliability. Feedback from the pilot test led to refinements in question wording and scale structure, such as adjusting ambiguous expressions

Table 1
Measurement and questionnaire.

Variable	Item	Source
Entertainment gratification	(1) I use short video applications to get relaxation and enjoy life.	[31]
	(2) I use short video applications to entertain myself.	
	(3) I can have fun anytime, anywhere by using short video applications.	
Information gratification	(1) By using short video applications, I am able to learn about the unknown.	[64]
	(2) I think short video applications are a good place to search for new knowledge.	
	(3) By using short video applications, I am able to learn useful things.	
	(4) By using short video applications, I am able to get new ideas.	
Sociality gratification	(1) I am able to meet new people through the use of short video applications.	[64]
	(2) By utilizing the short video application, I am able to connect with individuals who resonate with me.	
	(3) By using a short video application, I am able to talk to like-minded people.	
Immersive experience	(1) My sense of time easily slips away when I am engaged with short video applications.	[13]
	(2) I frequently find myself captivated while utilizing short video applications.	
	(3) When using short video applications, I often get so focused that I neglect other things.	
	(4) When using short video applications, I have felt that time passes quickly.	
Emotional attachment	(1) I am keen to use short video applications.	[45]
	(2) I believe I have an emotional connection with the short video applications.	
	(3) I have an emotional affinity towards the short video applications I use.	
	(4) I am genuinely attached to the short video applications.	
Compulsive short video app use	(1) I sometimes find it difficult to concentrate on my studies or work because of short video applications.	[63]
	(2) I have had less sleep due to spending more time on short video applications.	
	(3) Short video applications have affected my offline daily socializing activities.	
	(4) My family or friends used to believe that I dedicated an excessive amount of time to short video applications.	
	(5) I would experience anxiety in the absence of access to short video applications.	
	(6) I have tried to reduce the amount of time I spend using short video applications, but without success.	

and optimizing the logical flow of items. These procedures ensured that the translated measures accurately captured the constructs of interest and were suitable for the target Chinese population.

4. Data analysis strategy

This study performed statistical analyses using SPSS 24.0 and AMOS 24.0. The analyses incorporated controlled variables such as gender and age to ensure that extraneous influences were minimized, thereby enhancing the robustness of the findings. Data were preprocessed and cleaned in Microsoft Excel following a rigorous procedure, including dataset screening, validation, and verification to exclude incomplete responses. Additionally, anomalies and absent data points were detected and handled, while consistency checks were implemented. Subsequently, a comprehensive descriptive statistical analysis was performed, accompanied by an assessment of common method variance (CMV), utilizing the capabilities of SPSS. Moreover, to examine the interrelationships among the study’s factors, covariance-based structural equation modeling (SEM) was applied via AMOS. The process for

analyzing data employed a two-step methodology, involving a meticulous examination of the research model and its associated hypotheses. Firstly, confirmatory factor analysis was employed to test the constructed theoretical foundation, encompassing an assessment of the overall model appropriateness, construct reliability, and construct validity. Secondly, the proposed theoretical model underwent rigorous scrutiny through an examination of the structural equation model.

5. Results

5.1. Demographic statistics

Prior to delving into the proposed hypotheses, descriptive statistics derived from the survey responses were presented in Table 2. The data revealed that out of 896 participants, 458 (51.1 %) were male, while the remaining 438 participants (48.9 %) were female. The respondents encompassed individuals aged between 18 and 33 years, with a predominant proportion (45.3 %) falling within the 18 to 21 age brackets. In terms of educational attainment, the survey participants comprised 55.9 % undergraduate students, 41.9 % postgraduate students, and 2.2 % doctoral students. Regarding the utilization of short video apps, an overwhelming majority of individuals, accounting for 78.6 %, disclosed that they have been utilizing these apps for a duration exceeding three years. Moreover, approximately 75.9 % of the surveyed participants dedicated no less than three hours of their daily routine to engaging with short video apps.

5.2. Measurement model, reliability, and validity

As shown in Table 3, this measurement model was examined using confirmatory factor analysis (CFA) in the current study. The outcomes of the indicators assessing the fit of the estimation model ($\chi^2/\text{d.f.} = 2.485$; RMSEA = 0.027; RMR = 0.018; CFI = 0.953; AGFI = 0.962; IFI = 0.972; TLI = 0.964) reveal a satisfactory fit of the data. All metrics meet the criteria for acceptable model fit.

Table 4 delineates factor loadings, squared multiple correlations (SMC), average variance extracted (AVE), and composite reliability (CR) for the model's constructs. For the measurement model's reliability, CR assesses internal consistency among items. All five manifest variables examined in the present investigation exhibited CRs surpassing the criterion of 0.8, showing robust internal consistency throughout the constructs. Concurrently, the assessment of convergent validity through

Table 2
Summary of demographic information (N = 896).

Categories	Item	Frequency	Percentage (%)
Gender	Males	458	51.1
	Females	438	48.9
Age	18–21 years old	406	45.3
	22–25 years old	277	30.9
	26–29 years old	181	20.2
	30–33 years old	32	3.6
	Undergraduate degree	501	55.9
Educational background	Postgraduate degree	376	41.9
	Doctoral degree	19	2.2
Short video applications use experience	Less than 1 year	5	0.6
	1–2 years	21	2.3
	2–3 years	166	18.5
	3–4 years	299	33.4
	More than 4 years	405	45.2
Time spent on short video applications daily	Less than 1 h	11	1.2
	1–2 h	38	4.2
	2–3 h	167	18.6
	3–4 h	282	31.5
	More than 4 h	398	44.4

Table 3
Fit indices for the measurement model.

Model fit measures	Model fit criterion	Index value	Good model fit (Y/N)
Absolute fit indices			
RMSEA	<0.08	0.027	Y
RMR	<0.05	0.018	Y
$\chi^2/\text{d.f.}$ ($\chi^2 = 196.332$, d.f. = 79)	<3	2.485	Y
Incremental fit indices			
CFI	>0.9	0.953	Y
AGFI	>0.8	0.962	Y
IFI	>0.9	0.947	Y
TLI	>0.9	0.931	Y

Table 4
Statistical results of confirmatory factor analysis.

Constructs and items	Loading (>0.7)	SMC (>0.5)	CR (>0.7)	AVE (>0.5)
Entertainment gratification (EG)			0.831	0.621
EG1	0.798	0.637		
EG2	0.762	0.581		
EG3	0.803	0.645		
Information gratification (IG)			0.872	0.630
IG1	0.779	0.607		
IG2	0.818	0.669		
IG3	0.756	0.572		
IG4	0.821	0.674		
Sociality gratification (SG)			0.816	0.596
SG1	0.743	0.552		
SG2	0.782	0.612		
SG3	0.791	0.626		
Immersive experience (IE)			0.892	0.675
IE1	0.769	0.591		
IE2	0.833	0.694		
IE3	0.857	0.734		
IE4	0.825	0.681		
Emotional attachment (EA)			0.868	0.623
EA1	0.758	0.575		
EA2	0.811	0.658		
EA3	0.787	0.619		
EA4	0.799	0.638		
Compulsive short video app use (CU)			0.921	0.660
CU1	0.793	0.629		
CU2	0.789	0.623		
CU3	0.828	0.686		
CU4	0.805	0.648		
CU5	0.816	0.666		
CU6	0.843	0.711		

Notes: SMC, squared multiple correlations; CR, construct reliability; AVE, average variance extracted.

standardized load estimates, SMC, and AVE reveals noteworthy findings. All standardized load estimates surpass the threshold of 0.7, unequivocally signifying a robust manifestation of convergent validity across the measurement scale employed in this research. The SMC values exhibit a significant level of effectiveness, with the majority surpassing or closely approaching the 0.5 mark, thereby affirming the effectiveness of convergence. Additionally, all AVE values surpass the critical of 0.5, providing evidence of adequate convergence. By juxtaposing the square root of AVE for each construct with the squared correlation coefficients exhibited among the various constructs, an assessment of the discriminant validity of these variables is conducted. As depicted in Table 5, the square roots of the AVE for each construct surpass the inter-construct correlations, thereby substantiating the existence of discriminant validity.

Table 5
Discriminant validity.

	EG	IG	SG	IE	EA	CU
EG	0.788					
IG	0.476**	0.794				
SG	0.385**	0.121**	0.772			
IE	0.393**	0.355**	0.372**	0.822		
EA	0.417**	0.339**	0.306**	0.432**	0.789	
CU	0.289**	0.216**	0.257**	0.367**	0.386**	0.812

Notes: ** $P < 0.001$. EG, Entertainment gratification; IG, Information gratification; SG, Sociality gratification; IE, Immersive experience; EA, Emotional attachment; CU, Compulsive short video app use. Diagonal elements (bold) represent AVE. Off-diagonal elements represent squared correlations between variables.

5.3. Common method bias

Given that this study relies on a singular questionnaire-based approach for data collection, it is essential to acknowledge the potential presence of common method bias (CMB), which can arise due to the specific context and attributes of the project under investigation [65]. Common method bias denotes a systematic deviation stemming from the data gathering techniques employed in psychological or behavioral studies [66]. To identify the extent of CMB, the utilization of Harman's single-factor test is recommended. If a single factor emerges that accounts for the predominant portion of the covariance among the measurements, it can be inferred that a considerable degree of CMB exists. In the present article, the application of Harman's single-factor test was executed through the execution of exploratory factor analysis (EFA) on the comprehensive set of examined measurements, utilizing varimax rotation technique. The findings of the EFA unveiled that no discernible factors demonstrated a factor loading surpassing the designated criterion of 50 %, with the highest factor loading recorded at 27.1 %. Furthermore, the analysis contrasting multivariate with univariate frameworks within CFA indicates that CMV does not detract from the efficacy. The outcome serves as evidence that the current research is devoid of any concerns pertaining to CMB [67].

5.4. Evaluation of structure model

The adequacy of the structural model was assessed using AMOS 24.0. The assessment of the model's fit, as indicated by the $\chi^2/d.f.$ ratio of 1.826, demonstrates that the model satisfies the established criteria. Additionally, other fit indices provide evidence of a satisfactory model fit, including RMSEA = 0.022 < 0.05, RMR = 0.015 < 0.05, GFI = 0.961 > 0.9, AGFI = 0.973 > 0.9, NFI = 0.985 > 0.9, IFI = 0.988 > 0.9, TLI = 0.969 > 0.9, and CFI = 0.958 > 0.9.

Consistent with the hypothesized predictions, the regression analysis reveals that entertainment gratification ($\beta = 0.412$, $p < 0.001$) and information gratification ($\beta = 0.322$, $p < 0.001$) demonstrate substantial and statistically significant beneficial effects on immersive experience

(Fig. 2). These findings provide empirical evidence that both entertainment gratification and information gratification significantly predict users' immersive experience during short video app use. Consequently, the statistical validation of these relationships supports the confirmation of H1 and H2. Moreover, the results also confirm the positive effects of entertainment gratification ($\beta = 0.295$, $p < 0.001$), information gratification ($\beta = 0.218$, $p < 0.001$), and sociality gratification ($\beta = 0.316$, $p < 0.001$) on emotional attachment. These significant findings substantiate the validity of H4, H5, and H6, respectively. Furthermore, it is worth noting that the regression analysis reveals the significant positive impact of immersive experience ($\beta = 0.258$, $p < 0.05$) as well as emotional attachment ($\beta = 0.379$, $p < 0.01$) in shaping the emergence of compulsive use tendencies among users of short video apps. These findings provide robust support for the validation of H7 and H8. However, contrary to initial expectations, the analysis fails to establish a significant association between sociality gratification and immersive experience ($\beta = -0.122$, $p > 0.05$). Consequently, the empirical evidence contradicts H3. Following an exhaustive investigation into Hypotheses 1 to 8, this study delves into the potential mediating influences of immersive experience and emotional attachment concerning the link between gratifications sought and compulsive use of short video apps.

Bootstrapping analysis examined mediating relationships between gratifications sought and compulsive short video app use. Furthermore, the investigation employed bootstrapping methodologies to construct a confidence interval (CI) that would facilitate the estimation of the mediation effect, in accordance with the method delineated by academics [68]. The findings indicate that the relationship between entertainment gratification and compulsive behavior is mediated by two key factors: immersive experience (with a beta coefficient of 0.165, a statistically significant p-value of less than 0.05, specifically 0.001, and a 95 % confidence interval ranging from 0.078 to 0.390) and emotional attachment (evidenced by a beta coefficient of 0.162, a p-value of 0.001 which is again significant at the 0.05 level, and a confidence interval from 0.075 to 0.332). In a similar vein, the correlation between information gratification and the compulsive use of these apps is also partly mediated by the aforementioned factors of immersive experience ($\beta = 0.151$, $p = 0.001$, 95 % CI = [0.063, 0.271]) and emotional attachment ($\beta = 0.147$, $p = 0.001$, 95 % CI = [0.058, 0.252]). Additionally, the research delineates sociality gratification as an additional critical mediating variable in the nexus between emotional attachment and compulsive utilization of short video apps, as evidenced by a beta coefficient of 0.126, a p-value of 0.001, and a confidence interval extending from 0.051 to 0.177.

5.5. Multi-group invariance testing

Multi-group invariance (MGI) testing serves as an analytical framework that enables scholars to verify the uniformity of indicators within a measurement or structural model across multiple groups [69]. Predecessors in the field introduced a procedural model of invariance that delineated the measurement model in terms of factor loadings, factor

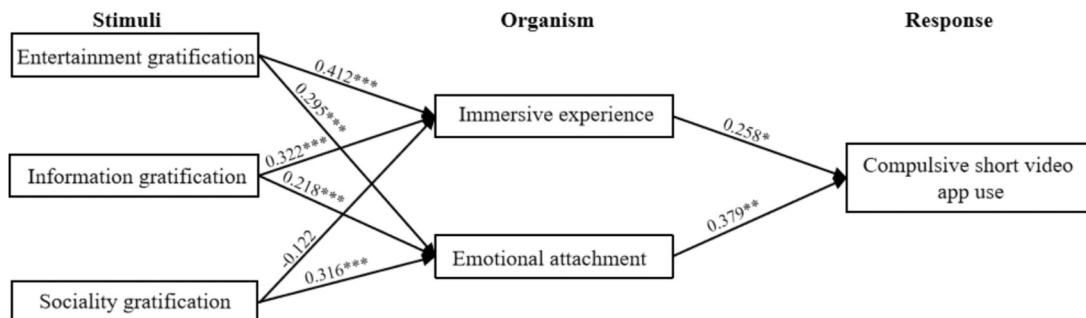


Fig. 2. Results for the structural equation model. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

correlations, and structural parameters elucidating hypothesized structural relationships. For the assessment of invariance, it was presupposed that these parameters remained constant across all groups. A non-significant chi-square difference test ($p > 0.05$) between the baseline and constrained models would suggest that the groups maintain invariance concerning these constraints [70]. Furthermore, even when the chi-square difference test indicates significant differences, maintaining the assumption of invariance may still be warranted if the changes in model fit parameters are minimal, typically below the threshold of 0.05 [71,70]. Grounded in these criteria, the research framework exhibits satisfactory invariance across factor loadings, factor correlations, and structural parameters among groups defined by gender, age, educational background, short video app use experience, and daily usage duration. Consequently, this highlights the integrity of the findings derived from the study. Table 6 provides a detailed presentation of the comprehensive results from the invariant analysis.

6. Discussion

6.1. Summary of the main findings

Based on the SOR framework, the primary objective of this study is to elucidate the underlying elements contributing to compulsive use of short video app among emerging adults. Specifically, it examines the interrelationships among gratifications sought, immersive experience, emotional attachment, and compulsive short video app use. The findings offer insights into the mechanisms underlying compulsive behaviors associated with short video apps. Perceived gratifications significantly shape users' immersive experience and emotional attachment, which in turn influence compulsive behaviors among younger users. Consequently, this study contributes to the understanding of these relationships, offering guidance for future development and optimization of short video apps. This cultural context of the inquiry, centered on a Chinese sample where platforms like Douyin are deeply integrated into daily life, undoubtedly provides fertile ground for investigating compulsive usage behaviors among short video app users. While the specific manifestations and prevalence of compulsive short video app use may differ between Eastern and Western users due to cultural distinctions, it is crucial to acknowledge that certain foundational elements identified in the research, such as the psychological effects of immersive experience and emotional attachment with digital content, could have broader applicability. These facets of human-digital interaction are presumed to reflect universal psychological mechanisms, as they are rooted in shared cognitive and affective processes that underpin individuals' engagement with media technologies.

Firstly, the findings for H1 and H2 confirm that entertainment gratification and information gratification are positive determinants of users' immersive experience. These results align with prior research on

short video app usage, which links immersive experience to users' pursuit of gratification [41,21,38]. Gratifications sought are capable of being perceived as users' subjective evaluations regarding the inherent characteristics of a digital application, subsequently influencing users' emotional responses to environmental stimuli[37,47]. Consequently, when users become deeply engaged in short video s, either experiencing enjoyment or seeking information, they exhibit heightened focus and are less likely to be distracted.

Contrary to previous research findings, the results for H3 in this study show no significant influence of sociality gratification on immersive experience. This finding offers fresh insights into the existing literature. Huang et al. have elucidated that the pursuit of sociality gratification on social media platforms can give rise to problematic usage patterns when it engenders excessively heightened levels of arousal, which constitutes an essential dimension of the flow state [38]. Nevertheless, this investigation reveals that users' perceived sociality gratification during the utilization of short video apps does not induce immersive experience. Immersion theory posits that individuals engage in activities for intrinsic enjoyment rather than external rewards [23]. Immersive experience generally demands a state of deep cognitive absorption, where individuals become deeply engrossed in an activity to the extent that their perception of temporal and spatial dimensions diminishes [8]. In contrast, social gratification often involves active, outward-facing engagement that seeks connections and interactions with others. This active engagement inherently requires awareness of one's social context and may evoke an emotionally vibrant state that is alert and connection-seeking. Therefore, the pursuit of social connections may inherently hinder the deep, introspective immersion that characterizes states of profound engagement. Users seeking social gratifications are often reminded of their physical and social environment as they seek feedback and interaction, which disrupts the absorption needed for immersive experience [41,72]. In summary, achieving immersion state often requires a high level of personal engagement and focus, which might be disrupted by the external focus required for social interaction. Consequently, this interpretation suggests that while social gratification on short video apps can provide meaningful and satisfying experiences, they do not typically promote the psychological detachment from one's immediate context that is necessary for immersive experience [40].

Secondly, the findings for H4-H6 validate the direct influence of entertainment gratification, information gratification, and sociality gratification on users' emotional attachment to short video apps. As users derive gratification from a platform that captivates their attention, short video apps are perceived as a valuable medium capable of augmenting pleasure and alleviating stress amidst their hectic routines, thereby fostering emotional attachment [47]. As users immerse themselves in video consumption and engage in interactions within the domain of short video apps, the resultant gratification they derive

Table 6
Multi-group invariance testing.

	Model Comparison	Df diff	CMIN diff	p-values	NFI diff	IFI diff	RFI diff	TLI diff
Gender	Unconstrained vs Measurement weights	15	23.302	0.035	0.007	0.006	-0.003	-0.003
	Measurement weights vs Structural weights	5	30.045	0.000	0.02	0.043	0.006	0.007
	Structural weights vs Structural covariance's	1	7.458	0.002	0.004	0.004	0.001	0.001
Age	Unconstrained vs Measurement weights	30	47.166	0.007	0.016	0.018	0.002	0.002
	Measurement weights vs Structural weights	10	36.762	0.000	0.024	0.022	0.005	0.006
	Structural weights vs Structural covariance's	2	2.155	0.297	0.002	0.002	0.000	0.000
Educational background	Unconstrained vs Measurement weights	15	20.672	0.118	0.006	0.007	-0.003	-0.003
	Measurement weights vs Structural weights	5	17.783	0.005	0.007	0.008	0.004	0.004
	Structural weights vs Structural covariance's	1	0.383	0.499	0.000	0.000	0.000	-0.002
Short video applications use experience	Unconstrained vs Measurement weights	30	36.336	0.177	0.013	0.015	-0.004	-0.005
	Measurement weights vs Structural weights	15	21.656	0.04	0.006	0.007	0.002	0.002
	Structural weights vs Structural covariance's	2	1.139	0.48	0.000	0.000	0.003	0.003
Time spent on short video applications daily	Unconstrained vs Measurement weights	30	58.851	0.002	0.011	0.025	0.004	0.004
	Measurement weights vs Structural weights	15	11.882	0.396	0.006	0.006	-0.002	-0.002
	Structural weights vs Structural covariance's	2	10.337	0.005	0.004	0.006	0.005	0.006

fosters a prolonged state of positive affect, thereby cultivating a favorable cognitive and affective perception of short video s. This heightened sense of pleasure serves to fortify their affiliation with mobile short video apps, consequently giving rise to emotional attachment and establishing extended patterns of usage behavior.

Thirdly, the findings pertaining to H7 and H8 reveal a positive association between immersive experience and emotional attachment with compulsive use of short video apps. These outcomes substantiate the proposition that both entertainment gratification and information gratification precipitate compulsive behavior by influencing individuals' immersive experience and emotional attachment to short video apps. Furthermore, this study confirms that immersive experience and emotional attachment mediate the relationship between gratifications sought and compulsive short video app use. Consistent with prior research, the present research provides evidence that the immersive experience and emotional attachment, as two nuanced psychological states engendered by users, substantially impact the emergence of compulsive behavior within the domain of short video apps [41,13,53]. The immersive nature of short video content enhances users' deep engagement, increasing their experienced gratification and fostering emotional attachment that blurs the line between volitional and habitual use. Both psychological states increase users' likelihood of engaging with short video apps, ultimately leading to compulsive usage behavior.

6.2. Theoretical and practical implications

This study advances compulsive behavior research by offering a diverse array of theoretical contributions, thereby deepening the conceptual understanding and enriching scholarly discourse. From a theoretical standpoint, this research builds on previous studies by examining the complex relationships between psychological motivation, online experiences, and the development of compulsive usage behavior in the context of short video apps. Firstly, despite previous research indicating that multiple factors contribute to compulsive usage of short video app, the conceivable pathway linking gratification, immersive experience, emotional attachment, and compulsive short video app use remains insufficiently explored. As a result, this empirical study provides an enunciated comprehension of the underlying mechanisms that drive the propensity towards compulsive utilization of short video app. Secondly, the article emphasizes the central role of psychological motivations in shaping compulsive behaviors and highlights the mediating roles of immersive experience and emotional attachment in this process. In particular, this research empirically illuminates the association between emotional states and compulsive use of short video app by substantiating the significant mediating influences manifested by immersive experience and emotional attachment. Thirdly, guided by the SOR paradigm, this study has formulated a conceptual research framework that unveils the nuanced process by which three distinct types of stimuli, mainly consisting of entertainment gratification, information gratification, and sociality gratification, elicit the younger generation's compulsive usage behavior on short video app. The framework also provides a detailed view of the antecedents and consequences of immersive experience and emotional attachment in the context of short video apps, enabling a comprehensive analysis of the factors influencing compulsive usage. This investigation enhances existing theoretical understanding of young users' actual behaviors on short video apps, while offering a new perspective to explore the latent mechanisms of compulsive use in this context.

From a practical perspective, these findings have important implications for service providers, young consumers, and the broader society associated with short video apps. Understanding which features are most valued by users allows app companies to enhance their effectiveness in promoting user engagement. Within this context, this paper advocates for the recognition of entertainment gratification, information gratification, and sociality gratification as motivational factors, and

encourages companies to invest efforts in refining the characteristics of their apps that foster the immersive experience and emotional attachment. For users, understanding companies' objectives and strategies can increase their awareness of potential risks and hazards associated with using short video apps. Many individuals, particularly young people and those with limited self-control, may struggle to reduce excessive reliance on these apps without external support. Consequently, educational institutions, colleges, and caregivers must exercise vigilance concerning the potentially adverse implications associated with these apps. By focusing on vulnerable users, this research provides valuable insights for parents, educators, and regulators about the factors that contribute to compulsive app usage. In the context of young adults being enticed by the vast array of entertainment content available on short video apps, it is crucial for educators and parents to inform them of the adverse consequences associated with app addiction and assist in establishing appropriate usage boundaries. For instance, educators and parents can institute sensible guidelines pertaining to the permissible duration of engagement with short video apps. Policymakers might consider regulations requiring digital platforms to incorporate features that promote digital well-being and mandate transparency regarding how algorithms operate. Additionally, these regulations may set restrictions on specific types of algorithmic content promotion that have been identified as encouraging addictive behaviors. For app designers, implementing features that promote mindful usage, such as usage reminders or duration trackers, could prove effective. For example, altering the algorithm to discourage prolonged binge sessions by intermittently suggesting breaks would be beneficial. By implementing measures such as monitoring or restricting interactions with certain features, platforms can assist users, especially the young generation, in mitigating compulsive behaviors. Furthermore, users should strive to bolster their self-regulatory capacities, or alternatively, platforms could integrate salient prompts to incentivize users to prioritize obligations over indulgence, thereby attenuating the exacerbating impact of procrastination on compulsive short video app use.

7. Limitations and implications for future research

Several limitations of the current work warrant explicit consideration in future investigations. Firstly, the study sample was restricted to young individuals aged 18 to 33 and limited to a single nation. While younger demographics dominate short video app usage, focusing solely on Chinese users may hinder the generalizability of findings, as cultural nuances in media consumption, social norms, and technological integration could shape compulsive behaviors differently across regions. Future research should therefore expand to include diverse age groups and cross-cultural comparisons, validating whether these foundational psychological mechanisms interact with cultural factors to influence compulsive short video app use. Secondly, this investigation primarily examines the influencing factors of compulsive usage through the lens of the SOR framework. However, it is noteworthy that non-cognitive factors, including individual habits and personalities, are not thoroughly accounted for in this analysis. Thus, future research undertakings should aim to delve into the realm of non-cognitive factors, thereby expanding the breadth of understanding in this domain. Thirdly, this study predominantly concentrates on the psychological variables contributing to compulsive usage behaviors associated with short video apps. Nonetheless, it acknowledges the significant role of technological aspects, including algorithmic content recommendations and app design features, in shaping user experiences and potentially exacerbating compulsive tendencies. Investigating how these technological factors interact with psychological variables in the future would provide a more holistic understanding of the forces driving compulsive usage. Furthermore, this study focuses solely on short video apps without comparing them to other forms of digital media. Future research could address this gap by examining cross-platform behaviors and identifying short video app attributes that exacerbate or mitigate compulsive use relative to

other digital media. Lastly, this research relies on cross-sectional analyses to explore the relationship between gratification and its effects on users. However, this cross-sectional approach is limited in establishing causal relationships within the theoretical framework, as short video app usage behaviors interact dynamically with evolving psychological and ecological variables over time. While findings indicate app engagement duration does not significantly differentiate users' propensity for compulsive use, this warrants further investigation to assess consistency across contexts and broader populations. Subsequent research could employ longitudinal tracking of users' interactions with short video apps to identify more nuanced temporal patterns and examine whether, and how rapidly, users transition from regular to compulsive usage.

CRedit authorship contribution statement

Hua Pang: Writing – original draft, Methodology, Investigation, Conceptualization. **Jingyuan Zhao:** Writing – original draft, Project administration, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] L. Lu, M. Liu, B. Ge, Z. Bai, Z. Liu, Adolescent addiction to short video applications in the mobile internet era, *Front. Psychol.* 13 (2022) 893599, <https://doi.org/10.3389/fpsyg.2022.893599>.
- [2] L.M. Meng, S. Kou, S. Duan, Y. Bie, The impact of content characteristics of Short-Form video ads on consumer purchase Behavior: evidence from TikTok, *J. Bus. Res.* 183 (2024) 114874, <https://doi.org/10.1016/j.jbusres.2024.114874>.
- [3] Y. Wang, Humor and camera view on mobile short-form video apps influence user experience and technology-adoption intent, an example of TikTok (DouYin), *Comput. Hum. Behav.* 110 (2020) 106373, <https://doi.org/10.1016/j.chb.2020.106373>.
- [4] X. Shen, J. Wang, How short video marketing influences purchase intention in social commerce: the role of users' persona perception, shared values, and individual-level factors, *Humanities and Social Sciences Communications* 11 (1) (2024) 290, <https://doi.org/10.1057/s41599-024-02808-w>.
- [5] N. Zhang, B. Hazarika, K. Chen, Y. Shi, A cross-national study on the excessive use of short-video applications among college students, *Comput. Hum. Behav.* 145 (2023) 107752, <https://doi.org/10.1016/j.chb.2023.107752>.
- [6] D. Qu, B. Liu, L. Jia, X. Zhang, D. Chen, Q. Zhang, R. Chen, The longitudinal relationships between short video addiction and depressive symptoms: a cross-lagged panel network analysis, *Comput. Hum. Behav.* 152 (2024) 108059, <https://doi.org/10.1016/j.chb.2023.108059>.
- [7] X. Wang, Q. Shang, How do social and parasocial relationships on TikTok impact the well-being of university students? the roles of algorithm awareness and compulsive use, *Acta Psychol.* 248 (2024) 104369, <https://doi.org/10.1016/j.actpsy.2024.104369>.
- [8] Y. Wang, S. Wang, Short-video applications use and self-concept clarity among adolescents: the mediating roles of flow and social media self-expansion, *Acta Psychol.* 249 (2024) 104469, <https://doi.org/10.1016/j.actpsy.2024.104469>.
- [9] Z. Yang, M.D. Griffiths, Z. Yan, W. Xu, Can watching online videos be addictive? a qualitative exploration of online video watching among chinese young adults, *Int. J. Environ. Res. Public Health* 18 (14) (2021) 7247, <https://doi.org/10.3390/ijerph18147247>.
- [10] L. Jia, F. Liang, The globalization of TikTok: strategies, governance and geopolitics, *Journal of Digital Media & Policy* 12 (2) (2021) 273–292, <https://doi.org/10.1386/jdmp.00062.1>.
- [11] S. Wang, N. Li, N. Liu, M. Habes, Transitioning from information sharing to knowledge services: unpacking the evolution of rural media within the context of media integration, *J. Knowl. Econ.* 1–32 (2023), <https://doi.org/10.1007/s13132-023-01593-2>.
- [12] K. Farhat, W. Aslam, I. Arif, Z. Ahmed, Does the dark side of personality traits explain compulsive smartphone use of higher education students? the interaction effect of dark side of personality with desirability and feasibility of smartphone use, *IIM Kozhikode Society & Management Review* 11 (1) (2022) 62–74, <https://doi.org/10.1177/22779752211000479>.
- [13] H.P. Lu, Y.H. Cheng, Sustainability in online video hosting services: the effects of serendipity and flow experience on prolonged usage time, *Sustainability* 12 (3) (2020) 1271, <https://doi.org/10.3390/su12031271>.
- [14] C. Peng, J.-Y. Lee, S. Liu, Psychological phenomenon analysis of short video users' anxiety, Addition and Subjective well-being, *International Journal of Contents* 18 (1) (2022), <https://doi.org/10.5392/IJoC.2022.18.1.027>.
- [15] P. Gerbaudo, TikTok and the algorithmic transformation of social media publics: from social networks to social interest clusters, *New Media Soc.* (2024), <https://doi.org/10.1177/14614448241304106>.
- [16] J. Lee, C. Abidin, T. Leaver, TikTok and children: an Introduction, *Int. J. Cult. Stud.* 28 (5) (2025) 893–906, <https://doi.org/10.1177/13678779251340419>.
- [17] C. Wang, M.K. Lee, Why we cannot resist our smartphones: investigating compulsive use of mobile SNS from a Stimulus-Response-Reinforcement perspective, *J. Assoc. Inf. Syst.* 21 (1) (2020) 4, <https://doi.org/10.17705/1jais.00596>.
- [18] J.E. Klobas, T.J. McGill, S. Moghavi, T. Paramanathan, Compulsive YouTube usage: a comparison of use motivation and personality effects, *Comput. Hum. Behav.* 87 (2018) 129–139, <https://doi.org/10.1016/j.chb.2018.05.038>.
- [19] S. Sanz-Blas, D. Buzova, M.J. Miquel-Romero, From Instagram overuse to instastress and emotional fatigue: the mediation of addiction, *Spanish Journal of Marketing-ESIC* 23 (2) (2019) 143–161, <https://doi.org/10.1108/sjme-12-2018-0059>.
- [20] S. Ponnusamy, M. Iranmanesh, B. Foroughi, S.S. Hyun, Drivers and outcomes of Instagram Addiction: Psychological well-being as moderator, *Comput. Hum. Behav.* 107 (2020) 106294, <https://doi.org/10.1016/j.chb.2020.106294>.
- [21] C. Gan, H. Li, Understanding the effects of gratifications on the continuance intention to use WeChat in China: a perspective on uses and gratifications, *Comput. Hum. Behav.* 78 (2018) 306–315, <https://doi.org/10.1016/j.chb.2017.10.003>.
- [22] J. Wan, Y. Lu, B. Wang, L. Zhao, How attachment influences users' willingness to donate to content creators in social media: a socio-technical systems perspective, *Inf. Manag.* 54 (7) (2017) 837–850, <https://doi.org/10.1016/j.im.2016.12.007>.
- [23] L. Wang, Y. Cui, J. Sun, J. Liu, D. Wei, C. Gu, Determinants of consumer adoption of multilingual self-service ordering systems in fast food restaurants, *Acta Psychol.* 245 (2024) 104216, <https://doi.org/10.1016/j.actpsy.2024.104216>.
- [24] T. Zhou, The effect of flow experience on users' social commerce intention, *Kybernetes* 49 (10) (2019) 2349–2363, <https://doi.org/10.1108/k-03-2019-0198>.
- [25] Y. Cheng, Y. Wang, W. Zhao, Shared Virtual reality Experiences during the COVID-19 Pandemic: Exploring the Gratifications and Effects of Engagement with Immersive Videos, *Int. J. Environ. Res. Public Health* 19 (9) (2022), <https://doi.org/10.3390/ijerph19095056>.
- [26] Y. Li, Y. Peng, What Drives Gift-giving Intention in Live Streaming? the Perspectives of Emotional Attachment and Flow Experience, *International Journal of Human-Computer Interaction* 37 (14) (2021) 1317–1329, <https://doi.org/10.1080/10447318.2021.1885224>.
- [27] N. Ahmad, H. Abdulkarim, The Impact of Flow Experience and Personality Type on the Intention to Use Virtual World, *International Journal of Human-Computer Interaction* 35 (12) (2019) 1074–1085, <https://doi.org/10.1080/10447318.2018.1509500>.
- [28] J.-É. Pelet, S. Ettis, K. Cowart, Optimal experience of flow enhanced by telepresence: evidence from social media use, *Inf. Manag.* 54 (1) (2017) 115–128, <https://doi.org/10.1016/j.im.2016.05.001>.
- [29] S.K. Lee, S.R. Min, Effects of information quality of online travel agencies on trust and continuous usage intention: an application of the SOR model, *The Journal of Asian Finance, Economics and Business* 8 (4) (2021) 971–982, <https://doi.org/10.13106/jafeb.2021.vol8.no4.0971>.
- [30] R. Qiao, C. Liu, J. Xu, Making algorithmic app use a virtuous cycle: Influence of user gratification and fatigue on algorithmic app dependence, *Humanities and Social Sciences Communications* 11 (1) (2024) 1–10, <https://doi.org/10.1057/s41599-024-03221-z>.
- [31] C.-L. Hsu, H.-P. Lu, Consumer behavior in online game communities: a motivational factor perspective, *Comput. Hum. Behav.* 23 (3) (2007) 1642–1659, <https://doi.org/10.1016/j.chb.2005.09.001>.
- [32] C. Xu, S. Ryan, V. Prybutok, C. Wen, It is not for fun: an examination of social network site usage, *Inf. Manag.* 49 (5) (2012) 210–217, <https://doi.org/10.1016/j.im.2012.05.001>.
- [33] A. Worrall, A. Cappello, R. Osolen, The importance of socio-emotional considerations in online communities, social informatics, and information science, *J. Assoc. Inf. Syst. Technol.* 72 (10) (2021) 1247–1260, <https://doi.org/10.1002/asi.24489>.
- [34] M.J. Kim, C.-K. Lee, M.W. Preis, Seniors' loyalty to social network sites: Effects of social capital and attachment, *Int. J. Inf. Manag.* 36 (6) (2016) 1020–1032, <https://doi.org/10.1016/j.jinfomgt.2016.04.019>.
- [35] M. Salehan, A. Negahban, Social networking on smartphones: when mobile phones become addictive, *Comput. Hum. Behav.* 29 (6) (2013) 2632–2639, <https://doi.org/10.1016/j.chb.2013.07.003>.
- [36] R.P. Yu, Y.W. Oh, Social media and expressive citizenship: Understanding the relationships between social and entertainment expression on Facebook and political participation, *Telematics Inform.* 35 (8) (2018) 2299–2311, <https://doi.org/10.1016/j.tele.2018.09.010>.

- [37] J. Guo, Z. Liu, Y. Liu, Key success factors for the launch of government social media platform: Identifying the formation mechanism of continuance intention, *Comput. Hum. Behav.* 55 (2016) 750–763, <https://doi.org/10.1016/j.chb.2015.10.004>.
- [38] L.-Y. Huang, Y.-J. Hsieh, Y.-C.-J. Wu, Gratifications and social network service usage: the mediating role of online experience, *Inf. Manag.* 51 (6) (2014) 774–782.
- [39] H. Khang, J.K. Kim, Y. Kim, Self-traits and motivations as antecedents of digital media flow and addiction: the internet, mobile phones, and video games, *Comput. Hum. Behav.* 29 (6) (2013) 2416–2424, <https://doi.org/10.1016/j.chb.2013.05.027>.
- [40] S. Miranda, I. Trigo, R. Rodrigues, M. Duarte, Addiction to social networking sites: Motivations, flow, and sense of belonging at the root of addiction, *Technol. Forecast. Soc. Chang.* 188 (2023), <https://doi.org/10.1016/j.techfore.2022.122280>.
- [41] K. Abrar, A.K. Mian, S. Zaman, How Social Gratification affects Social Network gaming Habitual Behavior. Sequential Mediation of Flow Experience and Consumer Satisfaction, *GMJACS* 12 (1) (2022) 112–128, <https://doi.org/10.59263/gmjacs.12.01.2022.270>.
- [42] F. Gao, Z. Shen, Sensory brand experience and brand loyalty: Mediators and gender differences, *Acta Psychol.* 244 (2024) 104191, <https://doi.org/10.1016/j.actpsy.2024.104191>.
- [43] F.R. Jiménez, K.E. Voss, An alternative approach to the measurement of emotional attachment, *Psychol. Mark.* 31 (5) (2014) 360–370, <https://doi.org/10.1002/mar.20700>.
- [44] X. Zhang, J. Choi, The importance of social influencer-generated contents for user cognition and emotional attachment: an information relevance perspective, *Sustainability* 14 (11) (2022) 6676, <https://doi.org/10.3390/su14116676>.
- [45] M. Thomson, D.J. MacInnis, C. Whan Park, The ties that bind: measuring the strength of consumers' emotional attachments to brands, *J. Consum. Psychol.* 15 (1) (2005) 77–91, https://doi.org/10.1207/s15327663jcp1501_10.
- [46] N. Choi, Information systems attachment: an empirical exploration of its antecedents and its impact on community participation intention, *J. Am. Soc. Inf. Sci. Technol.* 64 (11) (2013) 2354–2365, <https://doi.org/10.1002/asi.22939>.
- [47] Y. Xu, An exploration of the role played by attachment factors in the formation of social media addiction from a cognition-affect-conation perspective, *Acta Psychol. (Amst)* 236 (2023) 103904, <https://doi.org/10.1016/j.actpsy.2023.103904>.
- [48] X. Yin, J. Li, H. Si, P. Wu, Attention marketing in fragmented entertainment: how advertising embedding influences purchase decision in short-form video apps, *J. Retail. Consum. Serv.* 76 (2024) 103572, <https://doi.org/10.1016/j.jretconser.2023.103572>.
- [49] C. Wang, M.K.O. Lee, Z. Hua, A theory of social media dependence: evidence from microblog users, *Decis. Support Syst.* 69 (2015) 40–49, <https://doi.org/10.1016/j.dss.2014.11.002>.
- [50] L. Mi, T. Xu, Y. Sun, J. Zhao, T. Lv, X. Gan, L. Qiao, Playing Ant Forest to promote online green behavior: a new perspective on uses and gratifications, *J. Environ. Manage.* 278 (2021) 111544, <https://doi.org/10.1016/j.jenvman.2020.111544>.
- [51] R. Shi, M. Wang, C. Liu, N. Gull, The influence of short video platform characteristics on users' willingness to share marketing information: based on the SOR model, *Sustainability* 15 (3) (2023) 2448, <https://doi.org/10.3390/su15032448>.
- [52] L. Yang, X. Yuan, X. Yang, Study of the influencing mechanism of user interaction behavior of short video e-commerce live-streaming from the perspective of SOR theory and interactive ritual chains, *Curr. Psychol.* 43 (35) (2024) 28403–28415, <https://doi.org/10.1007/s12144-024-06494-4>.
- [53] X. Zhang, Y. Wu, S. Liu, Exploring short-form video application addiction: Socio-technical and attachment perspectives, *Telematics Inform.* 42 (2019), <https://doi.org/10.1016/j.tele.2019.101243>.
- [54] H. Zhang, Y. Lu, S. Gupta, L. Zhao, What motivates customers to participate in social commerce? the impact of technological environments and virtual customer experiences, *Inf. Manag.* 51 (8) (2014) 1017–1030, <https://doi.org/10.1016/j.im.2014.07.005>.
- [55] J. Berger, Word of mouth and interpersonal communication: a review and directions for future research, *J. Consum. Psychol.* 24 (4) (2014) 586–607, <https://doi.org/10.1016/j.jcps.2014.05.002>.
- [56] T.L. James, P.B. Lowry, L. Wallace, M. Warkentin, The effect of belongingness on obsessive-compulsive disorder in the use of online social networks, *J. Manag. Inf. Syst.* 34 (2) (2017) 560–596, <https://doi.org/10.1080/07421222.2017.1334496>.
- [57] O. Turel, R.D. Cavagnaro, Effect of abstinence from social media on time perception: differences between low-and at-risk for social media “addiction” groups, *Psychiatr. Q.* 90 (1) (2019) 217–227, <https://doi.org/10.1007/s11126-018-9614-3>.
- [58] D. Shin, F. Biocca, Exploring immersive experience in journalism, *New Media Soc.* 20 (8) (2018) 2800–2823, <https://doi.org/10.1177/1461444817733133>.
- [59] T. Friedrich, S. Schlauderer, S. Overhage, The impact of social commerce feature richness on website stickiness through cognitive and affective factors: an experimental study, *Electron. Commer. Res. Appl.* 36 (2019) 100861, <https://doi.org/10.1016/j.elerap.2019.100861>.
- [60] L. Littman, S. O'Malley, H. Kerschner, J.M. Bailey, Detransition and desistance among previously trans-identified young adults, *Arch. Sex. Behav.* 53 (1) (2024) 57–76, <https://doi.org/10.1007/s10508-023-02716-1>.
- [61] S.R. Rosenthal, S.L. Buka, B.D. Marshall, K.B. Carey, M.A. Clark, Negative experiences on Facebook and depressive symptoms among young adults, *J. Adolesc. Health* 59 (5) (2016) 510–516, <https://doi.org/10.1016/j.jadohealth.2016.06.023>.
- [62] E.J. Wolf, K.M. Harrington, S.L. Clark, M.W. Miller, Sample size Requirements for Structural Equation Models: an Evaluation of Power, Bias, and solution Propriety, *Educ. Psychol. Meas.* 73 (6) (2013) 913–934, <https://doi.org/10.1177/0013164413495237>.
- [63] S.B. Choi, M.S. Lim, Effects of social and technology overload on psychological well-being in young South Korean adults: the mediatory role of social network service addiction, *Comput. Hum. Behav.* 61 (2016) 245–254, <https://doi.org/10.1016/j.chb.2016.03.032>.
- [64] P. Patwardhan, J. Yang, H. Patwardhan, Understanding media satisfaction: Development and validation of an affect-based scale, *Atlantic Journal of Communication* 19 (3) (2011) 169–188, <https://doi.org/10.1080/15456870.2011.584508>.
- [65] H. Pang, Y. Ruan, Determining influences of information irrelevance, information overload and communication overload on WeChat discontinuance intention: the moderating role of exhaustion, *J. Retail. Consum. Serv.* 72 (2023) 103289, <https://doi.org/10.1016/j.jretconser.2023.103289>.
- [66] S. Bresciani, S.U. Rehman, G. Giovando, G.M. Alam, The role of environmental management accounting and environmental knowledge management practices influence on environmental performance: mediated-moderated model, *J. Knowl. Manag.* 27 (4) (2023) 896–918, <https://doi.org/10.1108/jkm-12-2021-0953>.
- [67] S.U. Rehman, A. Bhatti, S. Kraus, J.J. Ferreira, The role of environmental management control systems for ecological sustainability and sustainable performance, *Manag. Decis.* 59 (9) (2021) 2217–2237, <https://doi.org/10.1108/md-06-2020-0800>.
- [68] A. Alarifi, S. Bajaba, A. Basahal, From self-commitment to career crafting: the mediating role of career adaptability and the moderating role of job autonomy, *Acta Psychol.* 248 (2024) 104387, <https://doi.org/10.1016/j.actpsy.2024.104387>.
- [69] S. Adeel, A.D. Daniel, A. Botelho, The effect of entrepreneurship education on the determinants of entrepreneurial behaviour among higher education students: a multi-group analysis, *J. Innov. Knowl.* 8 (1) (2023) 100324, <https://doi.org/10.1016/j.jik.2023.100324>.
- [70] H. Pang, Y. Wang, W. Zhang, Can health information acquisition on mobile app influence psychological and physical well-being? Examining mediating role of bonding and bridging capital, *Acta Psychol.* 247 (2024) 104297, <https://doi.org/10.2139/ssrn.4707022>.
- [71] Y. Li, Y. Wang, Z. Ren, M. Gao, Q. Liu, C. Qiu, W. Zhang, The influence of environmental pressure on internet Use Disorder in adolescents: the potential mediating role of cognitive function, *Addict. Behav.* 101 (2020) 105976, <https://doi.org/10.1016/j.addbeh.2019.04.034>.
- [72] W. Oliveira, J. Hamari, Global Trends in Flow Theory Research within Gameful Environments: a Scoping Review, Bibliometric Analysis and Agenda for Future Studies (2024), <https://doi.org/10.24251/hicss.2023.163>.